(b) From your accurate titration results, calculate a suitable mean value to use in your calculations. Show clearly how you obtain the mean value.

25.0 cm^3 of **FB 1** required .... cm^3 of **FB 2**. [1]

(c) Calculations

(i) Give your answers to **(c)(ii)**, **(c)(iii)** and **(c)(iv)** to an appropriate number of significant figures. [1]

(ii) Calculate the amount, in mol, of manganate(VII) ions in the volume of **FB 2** in **(b)**.

amount of MnO_4^- = ... mol [1]

(iii) Use your answer to **(c)(ii)** and the equations at the start of the question to calculate the concentration, in $mol\,dm^{-3}$, of iron(II) ions in **FB 1**.

concentration of Fe^{2+} = ... $mol\,dm^{-3}$ [1]

(iv) Use your answer to **(c)(iii)** to calculate the concentration, in $g\,dm^{-3}$, of iron(II) ions in **FB 1**.

concentration of Fe^{2+} = ... $g\,dm^{-3}$ [1]

(v) The manufacturer of the iron supplement tablets used to make **FB 1** claims that each tablet contains a minimum of 150 mg of Fe^{2+}.

Use your answer to **(c)(iv)** and the information given about **FB 1** to determine whether this claim is correct. Show your working.

[1]

(d) A student used all the **FB 3** and suggests that dilute hydrochloric acid would be a suitable replacement.
Suggest whether the student is correct or not. Explain your answer.

....

....

.... [1]

[Total:14]